1. COMPONENTS LIST AND CONNECTION BLOCK DIAGRAM w/ PINS

Block diagram not applicable. Only Microchip Studio simulation was used.

2. SCHEMATICS

N/A

3. SCREENSHOTS OF EACH TASK OUTPUT (ATMEL STUDIO OUTPUT)

Keeping data from being overwritten was the biggest challenge by far of this design assignment. Below is a summary I needed to continuously reference to ensure I didn't mix anything up.

```
registers:
r0
r1
r2-r5           temp (eor segments, delta)
r6-r9           sum
r10-r13         y/z
r14
r15             eeprom management
r16             n (32x loop counter)
r17             i (8x loop counter)
r18-r21         eor products
r22-r25         load 1/4 of key at a time

other:

SRAM 0x100          v0
SRAM 0x104          v1
SRAM 0x108          delta
SRAM 0x200          encrypted message
PROM 0x400          raw message
PROM 0x600          key
EEPROM 0x00-0x10    key
```

Running the program reveals the following values

EEPROM:
Key

```
Memory: eeprom EEPROM        Address: 0x0000,eeprom
eeprom 0x0000  47 6f 6f 64 45 6e 6f 75 67 68 46 6f 72 4d 65 21 ff  GoodEnoughForMe!ÿ
eeprom 0x0011  ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff  ÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿ
```

General registers:
Note that R25-r22 correctly has the last bit of the key)

```
Registers
R00 = 0x00 R01 = 0x00 R02 = 0xB9 R03 = 0x79 R04 = 0x37 R05 = 0x9E R06 = 0x20 R07 = 0x37 R08 = 0xEF R09 = 0xC6 R10 = 0xA8 R11 = 0x15 R12 = 0x3E
  R13 = 0x34 R14 = 0x00 R15 = 0x21 R16 = 0x20 R17 = 0x08 R18 = 0x00 R19 = 0x38 R20 = 0x96 R21 = 0x45 R22 = 0x21 R23 = 0x65 R24 = 0x4D R25 = 0x72
  R26 = 0x10 R27 = 0x00 R28 = 0x40 R29 = 0x02 R30 = 0x40 R31 = 0x04
```

SRAM:
v0/v1/delta (Note that they are the expected values at the end of my program - last part of encrypted message as well as unchanged delta)

```
Memory: data IRAM        Address: 0x0100,data
data 0x0100  97 b6 74 32 34 3e 15 a8 9e 37 79 b9 00 00 00 00 00 00  —¶t24>.¨ž7y..
data 0x0112  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ..............
```

Encrypted message

```
Memory: data IRAM        Address: 0x0200,data
data 0x0200  63 ca b8 9f af 4c 58 ee 3f 85 c7 e4 7b a3 3c f0 ef 70  cÊ¸Ÿ¯LXî?.Çä{£<ðïp
data 0x0212  69 04 f4 4c de d3 18 b9 4c 16 2c a2 a8 de 98 98 dc 06  i.ôLÞÓ..L.,¢¨Þ˜˜Ü.
data 0x0224  e0 54 90 38 2f 82 05 15 1a ca 86 c1 be 18 d7 2f 3c 4d  àT.8/....Ê.Á..×/<M
data 0x0236  15 65 97 b6 74 32 34 3e 15 a8 00 00 00 00 00 00 00 00  .e—¶t24>.¨........
data 0x0248  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ..................
```

Program memory:

```
Memory: prog FLASH        Address: 0x0400,prog
prog 0x0400  54 68 69 73 20 69 73 20 6d 79 20 73 61 6d 70 6c 65 20  This is my sample
prog 0x0412  63 6f 64 65 20 66 6f 72 20 44 65 73 69 67 6e 20 41 73  code for Design As
prog 0x0424  73 69 67 6e 6d 65 6e 74 20 31 2e 20 50 65 72 66 65 63  signment 1. Perfec
prog 0x0436  74 6c 79 20 69 6e 74 61 63 74 ff ff ff ff ff ff ff ff  tly intactÿÿÿÿÿÿÿÿ
prog 0x0448  ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff  ÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿ
```

```
Memory: prog FLASH        Address: 0x0600,prog
prog 0x0600  47 6f 6f 64 45 6e 6f 75 67 68 46 6f 72 4d 65 21 ff ff  GoodEnoughForMe!ÿÿ
prog 0x0612  ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff  ÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿ
prog 0x0624  ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff  ÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿ
prog 0x0636  ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff  ÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿ
```

4. SCREENSHOT OF EACH DEMO (BOARD SETUP)

N/A